

Full length article

A Ti: LiNbO₃ optical waveguide sensor for measurement of intensive electric fieldJiahong Zhang^{*}, Qian Chen^a Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650500, China^b Yunnan Key Laboratory of Computer Technology Applications, Kunming 650500, China

ARTICLE INFO

Keywords:

Optical waveguide
Electric field sensor
Parameter design
Half-wave electric field
Power frequency electric field test

ABSTRACT

In this paper, a Ti: LN optical waveguide intensive electric field sensor without metal structure is designed and fabricated. By using the finite element simulation, the refractive index difference Δn_o and Δn_e of the waveguide are designed as 0.0105 and 0.0305 respectively when the prediffusion depth d is 5 μm , which results in the static operating point of the sensor is 86.4° when the length L is 45 mm and the prediffusion width w is 10 μm . The simulation results show that when the design of LN substrate is 6 mm (x) $\times$ 1 mm (y) $\times$ 45 mm (z), the ratio of the external and internal electric fields is 4.216, and the half-wave electric field is 10053 kV/m. The sensor is measured by the power-frequency electric field measurement system, and the half-wave electric field is about 10135 kV/m, and the maximum detectable electric field is about 3041 kV/m.

1. Introduction

Electric field sensor plays an important role in fields such as electric power and national defense, and are widely used for measuring lightning discharge electric fields, operation waves, fast transient over-voltages, electric field distributions near conductor surfaces of ultra-high-voltage (UHV) transmission lines, electric field distributions on the surfaces of insulators, electric field distributions inside high-voltage equipment, nuclear electromagnetic pulses, and high-power microwaves. In particular, with the implementation and application of UHV engineering, the voltage level continues to increase. Direct measurement of the power-frequency intensive electric fields near the transmission lines and the compact transmission line with split phase conductors has important significance for the design of transmission line and conductor arrangements, as well as the detection and analysis of the operating status of transmission lines and equipment. It should be noted that the peak electric field strength of these intensive electric fields can reach the order of MV/m [1,2], so researching an intensive electric field sensor that can measure several and even tens of MV/m electric fields will have significant application value.

The existing electric field sensors can be divided into traditional metallic antenan based electric field sensors and optical electric field sensors [3,4]. The former has the advantage of higher sensitivity, but the active circuit requires power supply, and the metallic structure will

interfere with the electric field under measurement, which results the maximum measurable electric fields of such sensor are usually several kV/m. In contrast, optical electric field sensors are passive devices, with small size, and wide bandwidth, making them suitable for measuring intensive electric fields and attracting the attention of researchers. Optical electric field sensors can be made using bulk electro-optic (EO) crystal or integrated optical waveguide (OWG). In 2015, Yang et al. developed a LN (LiNbO₃) bulk crystal based electric field sensor which has a maximum measurable electric field of 801 kV/m [5]. In 2021, Fan et al. developed a BGO (Bi₄Ge₃O₁₂) bulk crystal based electric field sensor suitable for monitoring AC/DC mixed electric field in oil medium environment, which can detect up to 1.67 MV/m AC electric field and 1.33 MV/m DC electric field [6]. In 2022, Zhang et al. developed a two-dimensional (2D) intensive electric field sensor by using one LN crystal, which can detect electric field up to 500 kV/m [7]. Due to the all-dielectric structure, such sensor has advantages of negligible interference to the electric field under measurement and is suitable for measurement of intensive electric field. However, as the requirement of discrete optical components such as the quarter-wave plate, the collimator, and the polarization analyzer etc., the complexity of the sensor structure is increased, which makes the optical path alignment is difficult and the stability is poor. In the 1980s, Bulmer et al. first reported an integrated OWG electric field sensor based on the Mach-Zehnder interferometer (MZI) structure at NRL [8], and subsequent researches are

^{*} Corresponding author at: Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650500, China.

E-mail address: zjh_mit@163.com (J. Zhang).

mainly focused on how to improve the sensitivity and bandwidth of the sensor [9–11]. To measure intensive electric field, in 2008, Celso et al. successfully developed a LN electrode-free OWG electric field sensor to detect intensive electric fields of up to 400 kV/m [12]. In 2012, Wang et al. developed an integrated OWG electric field sensor based on the common path interferometer (CPI) structure with metal electrode and antenna, which can detect electric field up to 840 kV/m [13]. In 2014, Zhang et al. developed a LN integrated OWG 3D electric field sensor using three MZI structures, which can be used to measure the lightning electromagnetic pulse (EMP) up to 370 kV/m [14]. In 2017, Celso et al. developed a LN electrodeless OWG electric field sensor with the maximum detectable peak electric field up to 200 kV/m [15]. In 2020, Zhang et al. designed a LN integrated OWG electric field sensor with segmented electrode, which can detect power frequency electric field up to 240 kV/m [16]. In 2022, Zhang et al. designed a LN integrated OWG direct current (DC) electric field sensor which can detect DC electric field up to 220 kV/m [17]. In 2023, Li et al. developed a LN integrated OWG electric field sensor by depositing a gold antenna array on the straight arm of the MZI for measuring the lightning EMP upto 200 kV/m [18].

It can be seen, to improve measuring sensitivity, the metallic antenna and electrode are usually designed and fabricated around the OWG, which results the sensor is not suitable for measurement of intensive electric field. In contrast, the electrodeless OWG sensor is more suitable for measuring the intensive electric field. But the maximum detectable electric field of existing electrodeless OWG sensor is no more than 1 MV/m.

In this article, for measuring the intensive electric field up to several MV/m, the waveguide parameters and half-wave electric field of an electrodeless OWG sensor are designed based on the finite element method (FEM). The sensor was developed using the titanium metal diffusion technology and finally a power frequency electric field measurement system was set up to test its half-wave electric field and other parameters.

2. Principles and structure of the sensor

The principle diagram of the Ti: LiNbO₃ OWG electric field sensor is shown in Fig. 1.

As shown in Fig. 1, the incident laser becomes linearly polarized light after passing through the polarizer. The linearly polarized light enters the LN OWG at an angle of 45° with the *x*-axis, and is divided into TE mode (polarized along the *x* direction) and TM mode (polarized along the *y* direction). When an electric field is applied along the *y* direction, based on the EO effect of the crystal, the laser transmitted along the *z* direction in the waveguide will be phase-modulated by the electric field,

and becomes elliptically polarized light carrying electric field information. The light then passes through a polarization-maintaining fiber (PMF) and enters a polarization analyzer. At this point, the polarization change is converted into optical intensity change, which is transmitted through a PMF to an optical detector, where it is converted into an electrical signal with information of the measured electric field. The relationship between the output optical power P_{out} and the input optical power P_{in} of the sensor is [19].

$$P_{out} = 0.5\alpha P_{in} [1 - b \cos(\varphi_B + \pi E_y / E_\pi)] \quad (1)$$

In Eq. (1), α is the optical power loss in the optical path, b is the extinction ratio of the sensor, E_π is the half-wave electric field of the sensor, E_y is the electric field applied along the *y* direction, and φ_B is the static bias point mainly determined by the waveguide structure. Here, φ_B can be expressed as

$$\varphi_B = \frac{2\pi}{\lambda} (n_{o-TE} - n_{o-TM})L \quad (2)$$

In formula (2), λ is the wavelength of input light, L is the length of EO interaction, n_{o-TE} and n_{o-TM} are the effective refractive indexes of TE mode and TM mode, respectively.

Because the relative dielectric constant of LN crystal is greater than the dielectric constant of air, the magnitude of the induced electric field on the waveguide will be smaller than the applied electric field. For k is the ratio of the external and the internal electric field of the sensor [20], then the half-wave electric field can be expressed as

$$E_\pi = \frac{k\lambda}{2(\frac{1}{2}n_{o-TE}^3 + \frac{1}{2}n_{o-TM}^3)\Gamma\eta\gamma_{22}L} \quad (3)$$

In Eq. (3), η is the antenna gain, $\Gamma = 0.2$ is the EO overlap factor, and $\gamma_{22} = 3.41 \times 10^{-12}$ m/V is the EO coefficient of the crystal. Since the sensor considered in this paper has no antenna or electrode structure, $\eta = 1$. Therefore, the half-wave electric field of the sensor is related to the optical wavelength, the length of the EO interaction, the refractive index of the waveguide mode, and the ratio of the external and internal electric fields.

It should be noted that when the electric field sensor is placed in an electric field, polarized charges will be generated on the surface of the LN crystal due to polarization effects, forming a polarization electric field E_p [21]. As the direction of the polarization electric field is opposite to the direction of the external electric field, the internal electric field strength in the waveguide can be expressed as

$$E_{int} = E_{ext} - E_p = E_{ext} - N_j \chi_j E_{int} \quad (4)$$

In Eq. (4), χ_j is the electrical polarizability of the LN crystal, N_j is the polarization factor of the LN crystal, and is related to the size of the LN

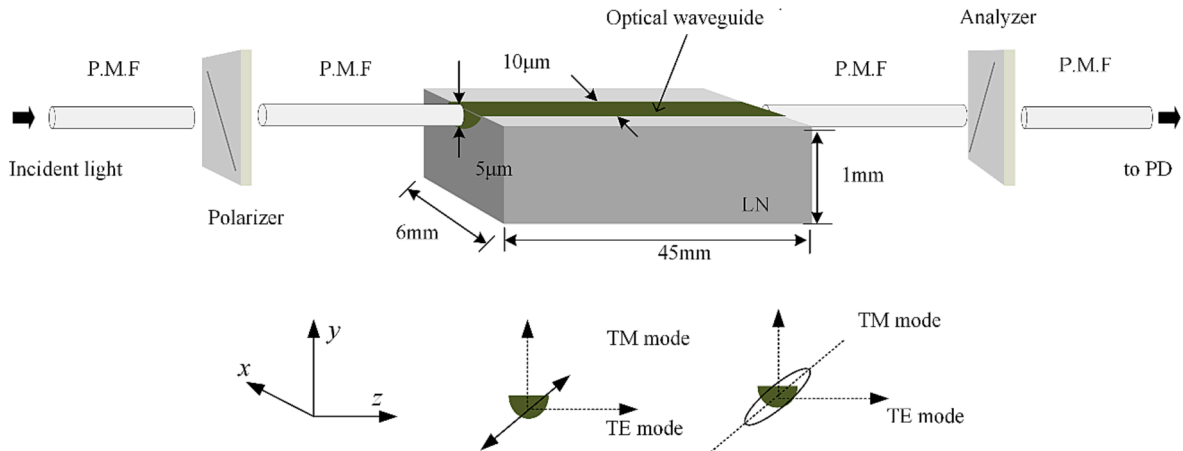


Fig. 1. Schematic diagram of the sensor.

crystal. The ratio coefficient k can be expressed as

$$k = \frac{E_{ext}}{E_{int}} = 1 + N_i \chi_j \quad (5)$$

According to Eqs. (3) and (5), it can be inferred that the half-wave electric field is also related to the size of the LN crystal.

3. Parameters design of the sensor

3.1. Design of the OWG parameters

In the design of OWG parameters, the main parameters of concern are the pre-diffusion width w , diffusion depth d , and refractive index increment Δn of the OWG. Firstly, the refractive index model of the waveguide is established to obtain the diffusion law of titanium metal in the LN crystal. When diffusing along the depth direction (along the y axis), the diffusion equation is given by

$$\frac{\partial C(y, t)}{\partial t} = \frac{\partial}{\partial y} \left[D(y, T) \frac{\partial C(y, t)}{\partial y} \right] \quad (6)$$

where, $C(y, t)$ represents the concentration distribution of the diffusion source, t is the diffusion time, and $D(y, T) = D_0 \exp(-E_p/K_b T)$ is the diffusion coefficient (E_p is the activation energy, $K_b = 8.62 \times 10^{-5} \text{ eV/K}$ is the Boltzmann constant, and T is the diffusion temperature). The diffusion coefficient D_0 is a constant. Typical values of the activation energy and D_0 can be obtained based on previous research results, as shown in Table 1.

In the actual production of waveguides, the refractive index of LN increases with the increasing concentration of Ti ions, and the refractive index distribution of the diffused OWG can be expressed as

$$n_i(x, y, \lambda) = n_{sub-i}(\lambda) + \frac{dn_i(\lambda)}{dC} C(y, t) \quad (7)$$

In Eq. (7), $i = o, e$, and $dn_o(\lambda)/dC = 0.33$, $dn_e(\lambda)/dC = 0.66$. $n_{sub, i}$ represents the refractive index of the LN substrate. When $\lambda = 1550 \text{ nm}$, according to reference [25], $n_{sub, o}$ and $n_{sub, e}$ can be calculated to be 2.2128 and 2.1375, respectively.

As shown in Eq. (7), the refractive index distribution of the waveguide layer is related to the concentration of Ti ions. Therefore, the concentration distribution of Ti ions $C(y, t)$ can be firstly obtained by solving Eq. (6), and then the refractive index distribution of the OWG can be further obtained.

It should be noted that the diffusion process of Ti ions into the LN substrate can be divided into two stages: firstly, the LN substrate surface is not completely diffused, and secondly, the diffusion source on the LN substrate surface is exhausted. Assuming that when the diffusion time is t_0 , the Ti film diffusion source on the crystal surface is exhausted, it can be analyzed that when $t \leq t_0$, the Ti concentration on the surface of LN can be maintained at a fixed value C , which is the saturation concentration of titanium ion in LN crystal. Therefore, the boundary conditions and initial conditions for diffusion are obtained as $C(0, t) = C$ and $C(y, 0) = 0$. In addition, for OWGs fabricated using titanium diffusion technology, the diffusion temperature is usually greater than 980°C [24], and the diffusion time is generally between 4~10 h, while the diffusion temperature should be kept below the Curie point (1125°C) of the LN crystal. Therefore, the diffusion temperature T is chosen to be 1050°C .

Using the COMSOL Multiphysics simulation software, the refractive

index distribution and mode field distribution of the Ti: LN OWG can be simulated by coupling the PDE (coefficients form partial differential equation) with the EWFD (electromagnetic wave frequency domain module). Fig. 2 shows relationship between the refractive index and diffusion depth along the $-y$ direction at different diffusion times. In this case, the diffusion depth d is equal to $|y|$.

Fig. 2(a) and Fig. 2(b) respectively describes the relationship between the ordinary and the extraordinary light refractive index with the diffusion depth at different diffusion times. It can be seen that the refractive index distribution of the waveguide depends on the diffusion time t . When the refractive index is constant, a longer diffusion time t will deepen the waveguide cross-section. At the same time, the increase in the waveguide cross-section will reduce the refractive index increment Δn and change the shape of the refractive index distribution curve. When the diffusion time t is 8 h, the Ti diffused LN ordinary light and extraordinary light refractive index distributions are shown in Fig. 3.

According to Fig. 3(a), the maximum and minimum values of ordinary light refractive index ($n_{o, max}$ and $n_{o, min}$) are 2.30 and 2.22, respectively. According to Fig. 3(b), the maximum and minimum values of extraordinary light refractive index ($n_{e, max}$ and $n_{e, min}$) are 2.32 and 2.14, respectively. Therefore, the increments $\Delta n_{o, max}$ and $\Delta n_{o, min}$ of the maximum and minimum values of the ordinary refractive index are 0.0172 and 0.0072, respectively. Similarly, the maximum and minimum values of the increment of extraordinary refractive index ($\Delta n_{e, max}$ and $\Delta n_{e, min}$) are 0.1625 and 0.0025, respectively. Schmidt and Kaminow found in 1974 that after titanium metal diffuses into LN crystal through high-temperature diffusion, both the ordinary refractive index and the extraordinary refractive index will increase, and $\Delta n_e > \Delta n_o$ ($\Delta n_e < 0.04$, $\Delta n_o < 0.02$) [26]. Therefore, when choosing the expected diffusion depth of $d = 5 \mu\text{m}$, $n_o \approx 2.2305$ and $n_e \approx 2.1705$ can be obtained from Fig. 2(a) and Fig. 2(b) respectively. Then, $\Delta n_e = 0.0305$ and $\Delta n_o = 0.0105$, and the increment of refractive index for both ordinary light and extraordinary light within the range of 0.02 to 0.04.

Based on the obtained refractive index distribution of the OWG, the optical wave modes in the waveguide were further simulated using finite element method. Fig. 4 shows the mode field distributions of TE mode and TM mode in the waveguide when the diffusion depth is $d = 5 \mu\text{m}$ and the diffusion width is $w = 12 \mu\text{m}$. The effective refractive indices of the TE mode and TM mode can be obtained as 2.2324 and 2.2322, respectively.

From Fig. 4, it can be seen that the mode field of both TE and TM modes are confined near the waveguide. By keeping the diffusion depth at $d = 5 \mu\text{m}$ and scanning the waveguide diffusion width w with parameterization, the relationships between $n_{eff}(\text{TE})$, $n_{eff}(\text{TM})$, birefringence ($n_{o, TE} - n_{o, TM}$), and w can be obtained for TE and TM modes, as shown in Fig. 5.

From Fig. 5, it can be observed that the birefringence ($n_{o, TE} - n_{o, TM}$) is related to the pre-diffusion width w of the waveguide. In order to ensure the sensor can operate in the nearly linear region, its optical bias point needs to be set as $\varphi_B = 2\pi (n_{o, TE} - n_{o, TM}) L / \lambda = 2m\pi + \pi/2$, where m is an integer [27].

Table 2 shows the values of the waveguide length L in the range of 40 mm to 55 mm and the corresponding pre-diffusion width w of the OWG in the range of 4 μm to 33 μm , which results in the optical bias point is set as $\varphi_B = 2m\pi + \pi/2 \pm 5^\circ$. Therefore, the optical bias point φ_B can be obtained by designing the pre-diffusion width w and length L of the OWG.

According to reference [28], for Ti: LN OWG, a pre-diffusion width of around 10 μm has good loss and transmission characteristics. Therefore, the pre-diffusion width of the OWG w is designed as 10 μm and the length L is designed as 45 mm in this paper, which results in φ_B is 86.4° .

3.2. Design of the process parameters

The Ti: LN waveguide parameters (Δn , d , w) are related to the initial width W_{Ti} and thickness τ of the titanium film. According to the research

Table 1
Typical values of activation energy and D_0

Reference	[22]	[23]	[24]
Tangential	y	y	y
$E_p(\text{eV})$	2.26	2.22	2.18
$D_0^y (\mu\text{m}^2/\text{h})$	1.5×10^8	1.09×10^8	7.9×10^8

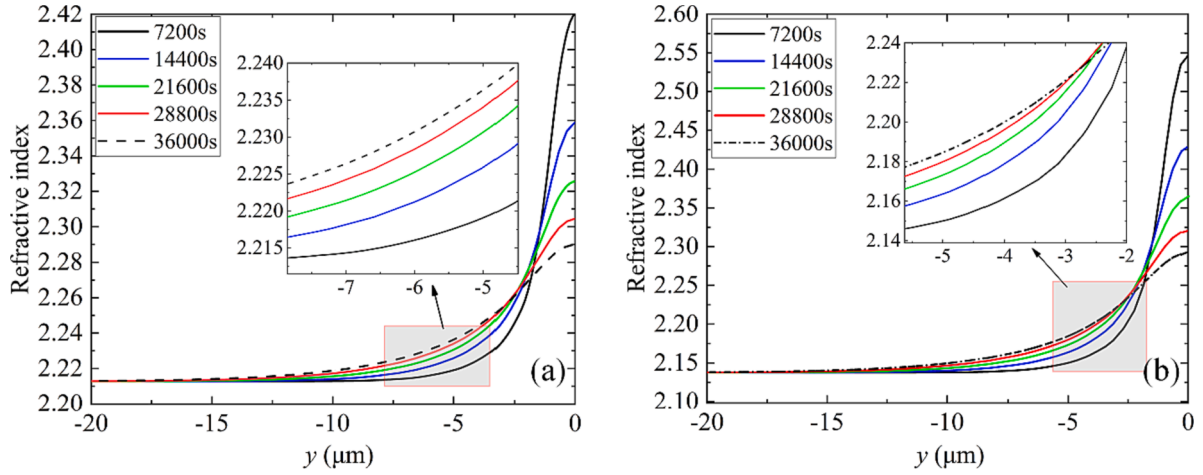


Fig. 2. Influence of diffusion time on refractive index distribution of (a) ordinary light and (b) extraordinary light.

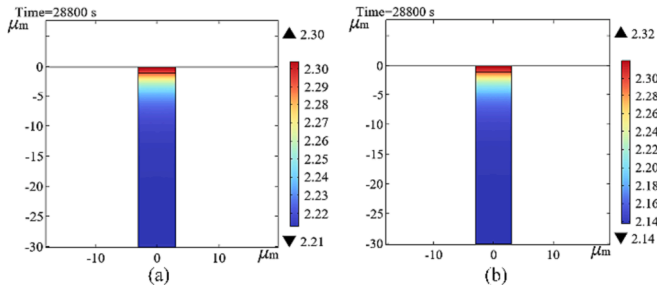


Fig. 3. Refractive index distribution of (a) ordinary light and (b) extraordinary light at $t = 8$ h.

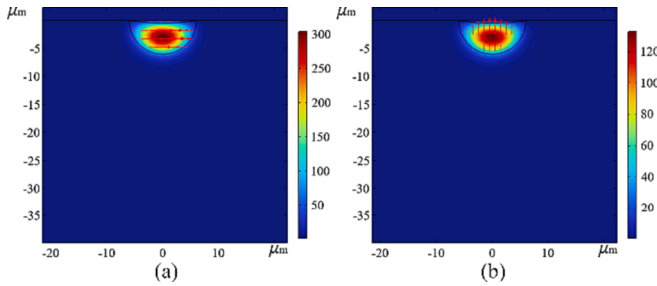


Fig. 4. Mode field distribution with wavelength of 1550 nm for (a) E_x component of TE mode and (b) E_y component of TE mode.

of C. Canali et al. [29], the change in ordinary refractive index of Ti: LN OWGs can be expressed as

$$\Delta n_o(\lambda) = \frac{dn_o(\lambda)}{dc} \frac{2\tau}{\sqrt{\pi}d} \quad (8)$$

When the diffusion depth $d = 5 \mu\text{m}$, $\Delta n_o = 0.0105$. By substituting into Eq. (8), we get the initial thickness τ of Ti film as 140 nm.

According to reference [30], the change of extraordinary refractive index Δn_e in Ti: LN OWG can be expressed as

$$\Delta n_e = \frac{dn_e}{dc} \frac{2}{\sqrt{\pi}} \alpha \frac{\tau}{d} \text{erf}\left(\frac{W_{Ti}}{w}\right) \quad (9)$$

In Eq. (9), $dn_e/dc = 7 \times 10^{-3}$ and $2\alpha/\pi = 0.225$ [29]. By substituting $d = 5 \mu\text{m}$, $\Delta n_e = 0.0305$, $\tau = 140$ nm and $w = 10 \mu\text{m}$ into Eq. (9), the initial width of Ti film can be obtained as $W_{Ti} \approx 6 \mu\text{m}$. In summary, we can obtain the designed waveguide parameters as shown in Table 3.

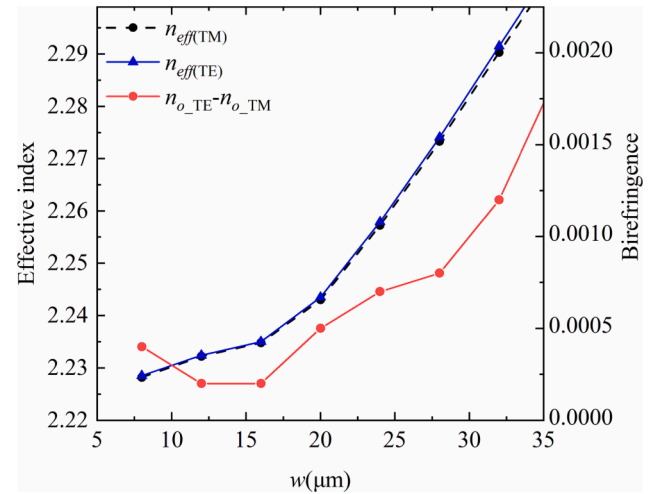


Fig. 5. The relationship between $n_{eff}(\text{TE})$, $n_{eff}(\text{TM})$, birefringence ($n_{o,\text{TE}} - n_{o,\text{TM}}$) and w .

Table 2

Values of w and L when $\varphi_B = 2\pi + \pi/2 \pm 5^\circ$.

$w(\mu\text{m})$	$L(\text{mm})$	φ_B	$w(\mu\text{m})$	$L(\text{mm})$	φ_B
4	49	85.0°	21	42	91.0°
6	48	91.8°	25	47	92.0°
7	46	86.4°	28	47	93.0°
9	53	92.0°	30	40/53	87.0°/93.6°
10	50/45	88.0°/86.4°	31	50	91.8°
19	45	94.5°	32	41/53	86.6°/92.9°
20	55	92.9°	33	54	92.0°

Table 3

Waveguide parameter.

Waveguide parameter					Ti-parameter	
	d	w	Δn_o	Δn_e	W_{Ti}	τ
values	5 μm	10 μm	0.0105	0.0305	6 μm	140 nm

3.3. Design of the LN substrate

To determine the geometrical dimensions of the LN substrate, a 3D simulation model of the Ti: LN OWG sensor is built using the COMSOL simulation software. The sensor was placed between two parallel copper

plates, and a mains voltage $U = 2000$ V with a frequency of 50 Hz is applied to the upper plate, while the lower plate is grounded, as shown in Fig. 6(a). Furthermore, the electric field strength distributions of the Ti: LN waveguide and the plates can be established, as shown in Fig. 6(b).

In the model shown in Fig. 6, the finite element domain is a sphere with a radius of 600 mm, the initial size of the LN substrate is 6 mm (x) $\times$ 1 mm (y) $\times$ 45 mm (z), the size of the two parallel copper plates is 0.35 m (long) $\times$ 0.35 m (wide) $\times$ 0.03 m (thick), and the spacing between the plates is $d_f = 75$ mm. The dimension of the plates meet the requirements of GB/T12720-1991 for the dimension specifications of a flat plate calibration system [31] (the spacing between parallel plates should be no less than 1.5 times the lateral dimension of the sensor, and the distance from any edge of the sensor to the bottom plate should be no less than $2d_f$). The specific material properties of each part in the simulation model are shown in Table 4.

During the simulation process, electric field probes are set at different positions (x, y, z) in the waveguide to monitor the change of the internal electric field E_{int} as a function of the LN substrate size. When the electric field probe is respectively placed at positions (0, $H/2-d/2$, 0), (0, $H/2-d/2$, -12.5), (0, $H/2-d/2$, 12.5), (0, $H/2-d/4$, 0), and (0, $H/2-d/2-d/4$, 0), with the width W of the LN substrate fixed at 6 mm and the length L fixed at 45 mm, the relationship between the substrate thickness H and the internal electric field E_{int} in the waveguide is shown in Fig. 7(a). When the substrate thickness H is fixed at 1 mm and the length L is fixed at 45 mm, the relationship between the substrate width W and the internal electric field E_{int} in the waveguide is shown in Fig. 7(b). When the substrate thickness H is fixed at 1 mm, the width W is fixed at 6 mm, and the relationship between the substrate length L and the internal electric field E_{int} in the waveguide is shown in Fig. 7(c).

According to the analysis in Fig. 7 (a), it can be found that when probes are set at different positions (x, y, z) in the Ti: LN OWG, the internal electric field E_{int} increased with the increase of substrate thickness H . When $0 \text{ mm} < H$ less than 3 mm, the internal electric field in the waveguide increased from 6062 V/m to 7147 V/m, showing an increase of 1085 V/m. When $H > 3$ mm, the internal electric field in the waveguide increased from 7147 V/m to 7631 V/m, showing an increase of 484 V/m.

According to the analysis in Fig. 7 (b), it can be found that when the probes are set at different positions (x, y, z) in the Ti: LN OWG, the internal electric field E_{int} decreased with the increase of substrate width W . When W is less than 6 mm, the internal electric field in the waveguide decreased sharply from 9014 V/m to 6840 V/m. When W greater than 6 mm, the change of internal electric field in the waveguide is relatively slow and eventually tends to a constant value.

According to the analysis in Fig. 7(c), it can be found that when the probe is set at position (0, y , 0) in the waveguide, the internal electric field in the waveguide remains basically unchanged with LN crystal length ranging from 10 mm to 60 mm. When the probe is set at position (0, y , z) in the waveguide, the internal electric field in the waveguide

Table 4
Material properties of each part.

Simulation parts	Materials	Refractive index	Electrical conductivity (s/m)	Relative magnetic permeability
EO crystal	LN	n_o	4.719×10^{-14}	0
		n_e	3.776×10^{-14}	0
Waveguide	Ti: LN	$n_o + \Delta n_o$	5.077×10^{-14}	0
		$n_e + \Delta n_e$	4.086×10^{-14}	0
Parallel plates	Cu	1	5.998×10^7	1
Finite element domain	air	1	0	1

remains basically unchanged with LN substrate length ranging from 40 mm to 60 mm. Further analysis shows that when the probes are set at different positions (x, y, z) in the waveguide, the changes of internal electric fields in the waveguide are very slow and eventually tends to a constant value when the substrate length L varied within the range of 40 mm to 60 mm.

In summary, as shown in Fig. 7, the internal electric field E_{int} of the Ti: LN OWG electric field sensor is indeed related to the size of the LN substrate, that is, k is related to the size of the LN substrate. In order to adapt to the measurement of high field strength, the value of k should be reasonably designed to make the designed sensor have a sufficiently large half-wave electric field, which requires the value of the internal electric field E_{int} in the waveguide to be as small as possible. In this paper, the thickness H of the LN substrate is designed to be 1 mm, the width W is 6 mm, and the length L is 45 mm. By adjusting the voltages applied on the two parallel plates electrodes (1.5~14.0 kV according to the experimental conditions) to generate uniform electric fields of different strengths, and using electric field probes at different positions (x, y, z) in the waveguide to detect the internal electric fields, the relationship between the external electric fields E_{ext} and the internal electric fields E_{int} in the waveguide are obtained, when the electric field probes are respectively placed at (0, $H/2-d/2$, 0), (0, $H/2-d/2$, -12.5), (0, $H/2-d/2$, 12.5), (0, $H/2-d/4$, 0), (0, $H/2-d/2-d/4$, 0). The results are shown in Fig. 8.

From Fig. 8, it can be concluded that when the probes are set at different positions (x, y, z) in the waveguide, the ratios between the external electric field and the internal electric field in the waveguide is approximately 4.216. Further substituting the wavelength $\lambda = 1550$ nm, $n_{o\text{TE}} = 2.2233$, $n_{o\text{TM}} = 2.168$ into formula (3), the half-wave electric field E_{π} can be calculated to be 10053 kV/m.

4. Sensor fabrication and measurement

4.1. Fabrication of the sensor

According to the parameters given in Table 3, an electric field sensor with dimensions of 65 mm (z) $\times$ 20 mm (x) $\times$ 7 mm (y) is fabricated and packaged. The fabrication process of the Ti: LN OWG is shown in Fig. 9. A y -cut and z -propagation LN crystal is used as the substrate, thoroughly cleaned and dried, and then coated with photoresist on its surface, followed by drying and cooling. After masking, exposure, developing, and fixing, a titanium film with the required thickness is evaporated by electron beam and then the photoresist is removed and stripped off in acetone solution, leaving the required metal titanium strip. Finally, the Ti: LN OWG device is formed by being sent into a diffusion furnace and maintaining the diffusion temperature at 1050°C for 8 h. The photograph of the ceramic packaged electric field sensor is shown in the Fig. 10.

4.2. Testing of the sensor

According to the schematic diagram of the power frequency electric field test system, shown in Fig. 11(a), an experimental system is set up to

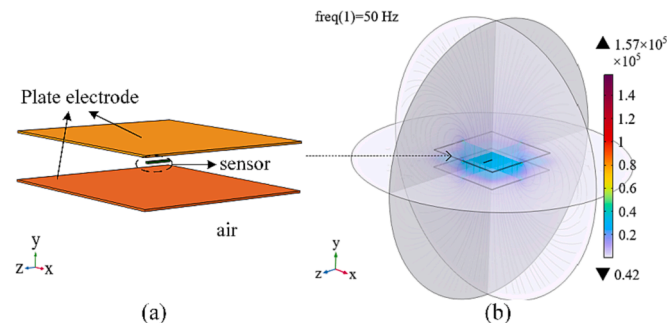


Fig. 6. Simulation of the sensor. (a) Simulation model. (b) Electric field distribution.

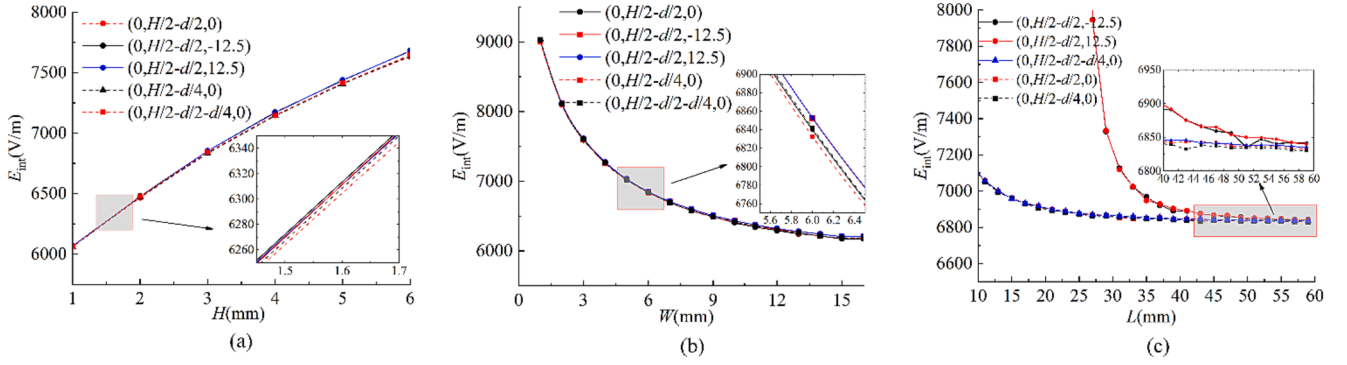


Fig. 7. Relation between E_{int} and size of LN. (a) E_{int} versus H . (b) E_{int} versus W . (c) E_{int} versus L .

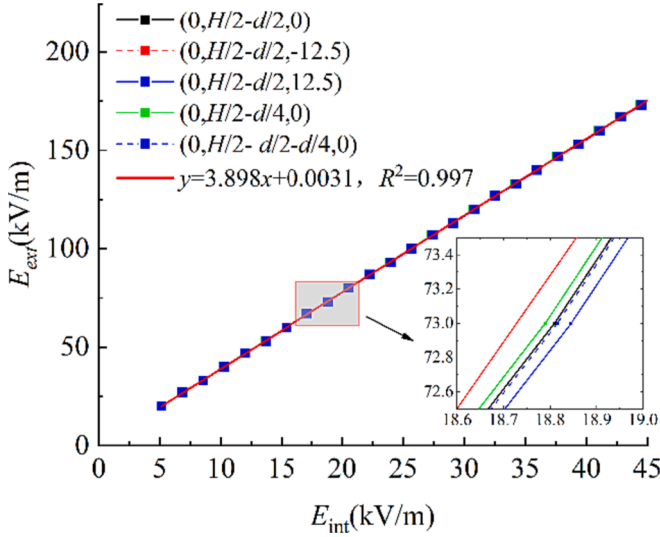


Fig. 8. Variation curve between the internal and external electric fields of the sensor.

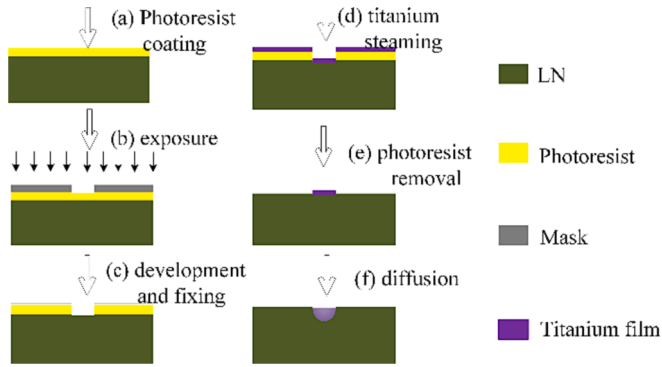


Fig. 9. Fabrication process of Ti: LN OWG.

test the input/output characteristics of the sensor. The copper electrode plate used in the experimental system has a size of $0.35 \text{ m} \times 0.35 \text{ m} \times 0.03 \text{ m}$, and the distance between the two electrode plates is 75 mm . The input end of the power control box is connected to the $220 \text{ V}/50 \text{ Hz}$ mains power supply, and the output end is connected to the high voltage transformer to boost the voltage. The transformer output high voltage is connected to two parallel copper plates, generating an intensive electric field between the two plates. By adjusting the control box knob to stabilize the output peak voltages of the transformer from 1 kV to 14 kV , results the $13\text{--}187 \text{ kV/m}$ uniform electric fields are generated between

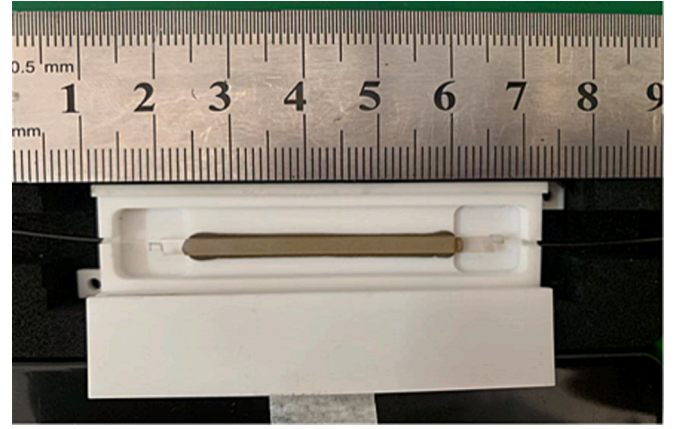


Fig. 10. Photograph of the sensor after fabrication.

the two plates.

The output light of the tunable optical source is connected to the input of the sensor through PMF, and the output of the sensor is connected to the photo receiver through PMF. After photoelectric conversion, the photo receiver outputs an electrical signal, which is observed and measured using an oscilloscope with bandwidth of 200 MHz and sampling rate of 2 GS/s . Fig. 12 shows the output waveforms of the sensor system when the applied electric fields are 27 kV/m , 53 kV/m , 133 kV/m , and 186 kV/m , respectively. It can be seen from Fig. 12 that the sensor can detect the temporal waveform of external electric fields without distortion.

Further changing the applied electric field strength, the input-output characteristic of the sensor is measured and shown in Fig. 13. Using linear fitting $y = ax + b$, the linear correlation coefficient $R^2 = 0.9826$ and the sensitivity of the sensor is $0.012 \text{ mV}/(\text{kV/m})$ are obtained. It should be noted that the noise floor of the sensor system is 2.3 mV , and under the condition that the signal is higher than the noise by 6 dB , the minimum measurable electric field of the sensor is approximately 192 kV/m (E_{min}).

Due to the half-wave electric field of the sensor is larger, and the maximum electric field that can be generated under existing experimental conditions is limited, the maximum linear measurable electric field cannot be measured directly. According to Eq. (1), the input-output relationship of the measurement system can be expressed as

$$V_{out} = 0.5\alpha A P_{in} [1 - b \cos(\varphi_B + \pi E_y/E_\pi)] \quad (10)$$

In Eq. (10), A is the photoelectric conversion coefficient of the photodetector. When $\pi E_y/E_\pi \ll 1$, Eq. (10) can be further simplified as follows

$$V_{out} \approx 0.5\alpha A P_{in} b \pi E_y/E_\pi + 0.5\alpha A P_{in} \quad (11)$$

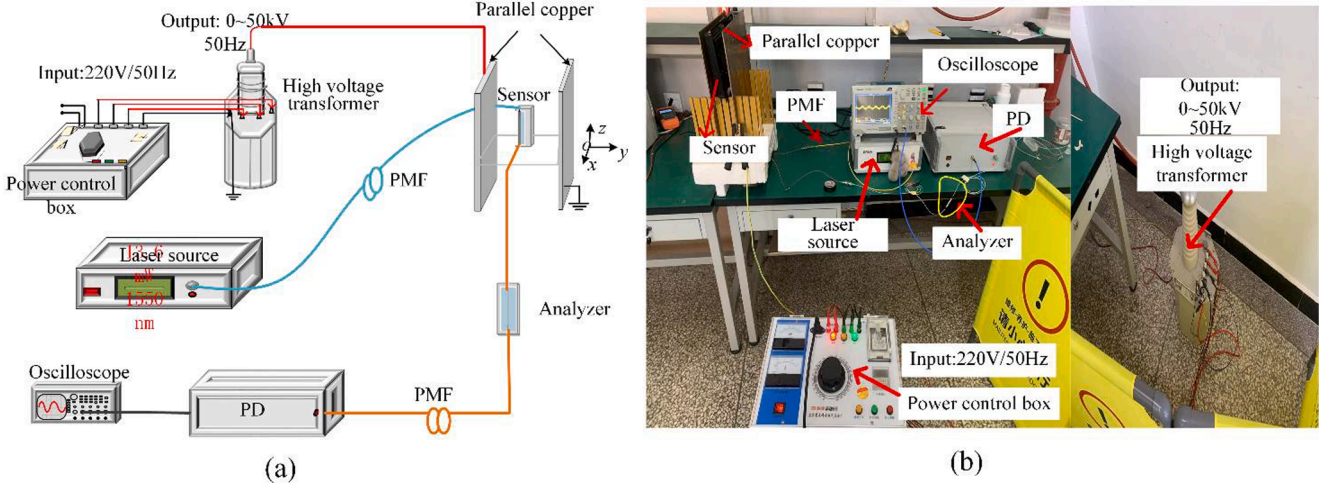


Fig. 11. Power frequency electric field test. (a) Diagram of the measurement system. (b) Photos of the measurement system.

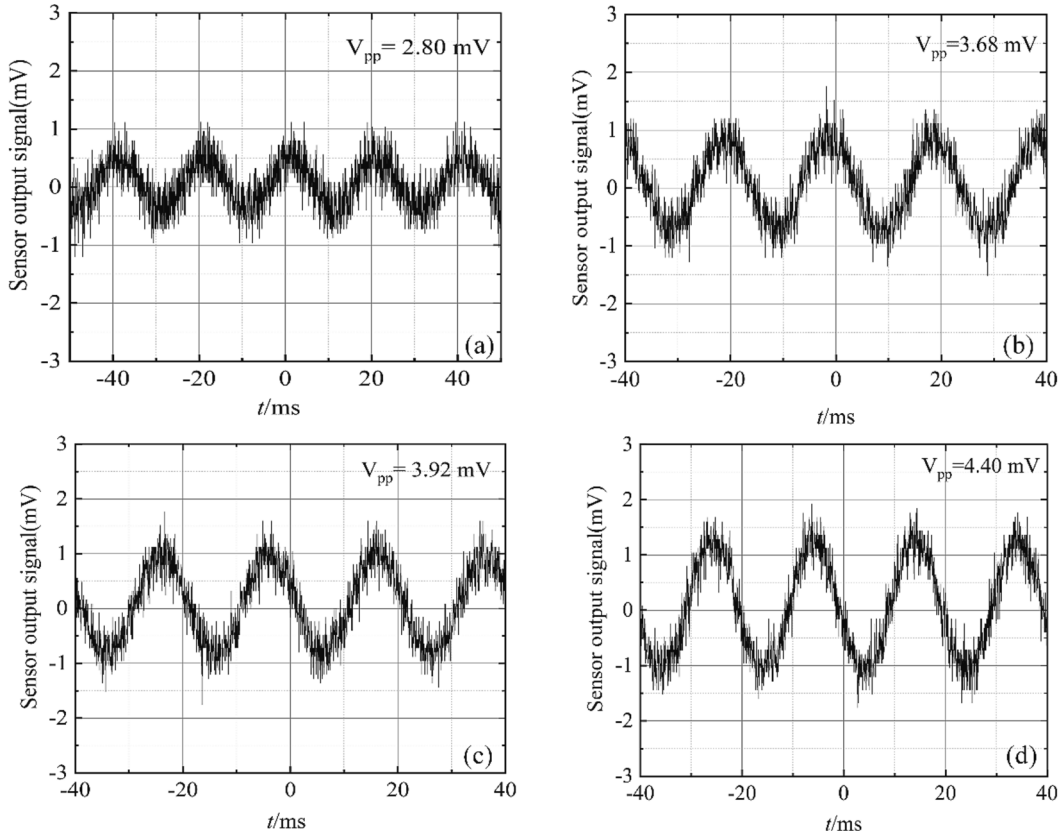


Fig. 12. Output waveforms of the sensor system under different fields. (a) $E = 53$ kV/m (b) $E = 106$ kV/m (c) $E = 133$ kV/m (d) $E = 186$ kV/m.

In experiment, $\alpha P_{in} = 108 \mu W$, and according to the linear fitting results, $A = 745.936$ mV/mW, then for $b = 1$ are taken and ignoring the DC component, according to Eq. (11), $E_{\pi} \approx 10135$ kV/m can be obtained, which is basically consistent with the theoretical calculation of the half-wave electric field value. According to reference [16], the maximum linear measurable electric field of the sensor is about $0.3 E_{\pi}$, that is, the maximum measurable electric field is approximately 3041 kV/m.

Further substituting the output voltage of the sensor system into the linear fitting line $y = 0.012x + 2.3$, the solved electric field strength is taken as the measurement value of the system and denoted as E_{M-i} . At the same time, according to $E_{T-i} = U/D$, the electric field strength is calculated as the true value (where U and D represent the applied voltage and

electrode plates spacing, respectively). Then, the root mean square error $RMSE$ can be expressed as follows

$$RMSE = \sqrt{\frac{1}{M} \sum_{i=1}^M (E_{T-i} - E_{M-i})^2} \quad (12)$$

In Eq. (12), M represents the number of measurement times. The measurement and true values are shown in Table 5.

By substituting the data in Table 5 into Eq. (12), the $RMSE$ is calculated as 9.849. However, consider the error of the high-voltage transformer output voltage is about $\pm 1.5\%$, which results in the $RMSE$ of the sensor system changes from 10.648 to 9.411. In addition, consider the measurement error caused by the unstable output optical

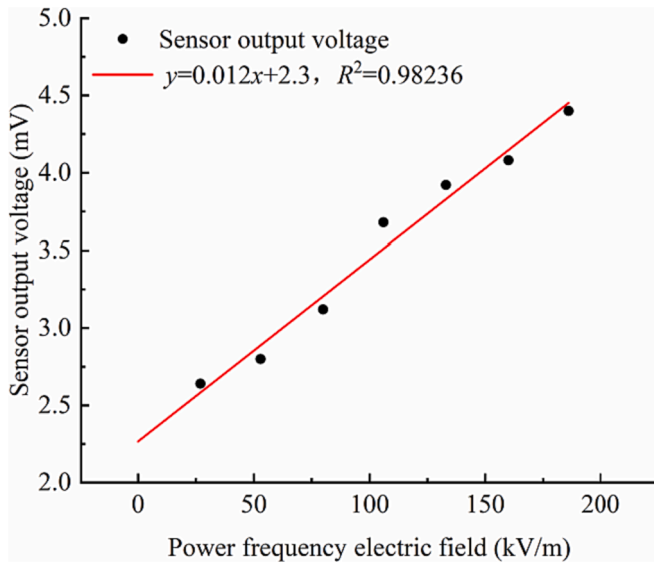


Fig. 13. Input/output characteristics of the sensor.

Table 5

The measured and true values of the electric fields.

<i>i</i>	1	2	3	4	5	6	7
E_{M-i} (kV/m)	35	42	68	115	135	148	175
E_{T-i} (kV/m)	27	53	80	106	133	160	186

power of the DFB light source is [32]

$$\varepsilon_{error} = |P_{out} - P'_{out}| / P_{out} \propto \Delta P_{in} / P_{in} \quad (13)$$

where, ΔP_{in} is the variation of the laser output optical power. For the used DFB light source, the $\Delta P_{in} / P_{in}$ is about 3% through multiple measurements, which results in the *RMSE* is 8.944. Therefore, both the errors of the high-voltage transformer output voltage and the output unstable power of the DFB light source can affect the measurement error of the sensor system.

Moreover, the performance comparison of similar optical electric field sensors has been shown in Table 6. It can be seen from Table 6 that the BGO bulk crystal based electric field sensor has the maximum detectable electric field of 1.67 MV/m. However, as some discrete optical components including the 1/4 wave plate, the collimator, and the polarization analyzer etc. are required, the complexity of the sensor structure is increased, which makes the optical path alignment is difficult and the stability is poor. However, the proposed LN OWG electrodeless electric field sensor in this paper has advantages of compact structure and larger measurable electric field, which is more suitable for the measurement of intensive electric field.

5. Conclusion

In this paper, the COMSOL simulation software is used to simulate and design the OWG parameters, the size of the LN substrate, and the half-wave electric field. Finally, the half-wave electric field is designed to be 10053 kV/m. Based on the designed parameters, an OWG intensive electric field sensor is fabricated with the encapsulated size is 65 mm × 20 mm × 7 mm. A power frequency electric field testing system is set up to test the sensor, and the half-wave electric field is found to be about 10135 kV/m, results in the maximum measurable electric field is about 3041 kV/m. The sensor can be used to measure intensive electric fields without any metal structure, avoiding interference with the measured electric field. However, due to the limitation of experimental

Table 6

Performance comparison of similar optical electric field sensors.

Reference	Configuration	Dimensions	Electrode	1/4 wave plate, etc.	E_{max}
[7]	LN bulk crystal	87 × 20 × 20 mm ³	no	yes	500 kV/m
[6]	BGO bulk crystal	20 × 8 × 8 mm ³	no	yes	1.67 MV/m
[17]	LN OWG	40 × 5 × 1 mm ³	yes	no	220 kV/m
[13]	LN OWG	50 × 12 × 5 mm ³	yes	no	840 kV/m
This work	LN OWG	45 × 6 × 1 mm ³	no	no	3.04 MV/m

conditions, actual test of the maximum detectable electric field was not carried out. In fact, even if the experimental conditions are met, unexpected problems may be encountered in the actual testing and verification. For example, when the test electric field reaches the MV/m level, due to the influence of ambient temperature, humidity, pressure, etc., air breakdown is very likely to occur, posing a threat to the normal operation of the test system and the safety of personnel. In addition, under such a large electric field, the resonance caused by the inverse piezoelectric effect of LN crystal may also affect the measured electric field. Therefore, for the actual test of larger electric field, all these problems need to be further considered and studied in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China under Grant 62162034, and in part by Yunnan Fundamental Research Projects under Grant 202201AT070189.

References

- [1] H. Kirkham, Measurement of electric fields generated from alternating current, *IEEE Instrum. Meas. Mag.* 9 (2006) 58–61, [10.1109/2FMIM.2006.1708355](https://doi.org/10.1109/2FMIM.2006.1708355).
- [2] K. Hidaka, H. Fujita, A new method of electric field measurements in corona discharge using Pockels device, *J. Appl. Phys.* 53 (1982) 5999–6003, <https://doi.org/10.1063/1.331568>.
- [3] W. Feser, W. Pfaff, A potential free spherical sensor for the measurement of transient electric field, *IEEE Trans. PAS.* 103 (1984) 2904–2911, <https://doi.org/10.1109/TPAS.1984.318292>.
- [4] D.J. Lee, N.W. Kang, J.H. Choi, J.F. Whitaker, Recent advances in the design of electro-optic sensors for minimally destructive microwave field probing, *Sensors* 11 (2011) 806–824, <https://doi.org/10.3390/s110100806>.
- [5] Q. Yang, S. Sun, R. Han, W. Sima, T. Liu, Intense transient electric field sensor based on the electro-optic effect of LN, *AIP Adv.* 5 (2015) 107–130, <https://doi.org/10.1063/1.4934720>.
- [6] X. Fan, W. Chen, C. Guo, Z. Wu, Sensitivity characteristics of EO sensors under AC/DC hybrid electric fields in oil medium environment *IEEE Trans. Instrum. Meas.* 70 (2021), <https://doi.org/10.1109/TIM.2021.3096272>.
- [7] Y. Zhang, J. Zhang, Y. Li, H. Lei, An optical intense 2D electric field sensor using a single LN crystal, *Curr. Opt. Photon.* 6 (2022) 183–190, <https://doi.org/10.3807/COPP.2022.6.2.183>.
- [8] C.H. Bulmer, W.K. Burns, R.P. Moeller, Linear interferometric waveguide modulator for electromagnetic-field detection, *Opt. Lett.* 5 (1980) 176–178, <https://doi.org/10.1364/OL.5.000176>.
- [9] B. Sun, F. Chen, K. Chen, Z. Hu, Y. Cao, Integrated optical electric field sensor from 10 kHz to 18 GHz, *IEEE Photon. Technol. Lett.* 24 (2012) 1106–1108, <https://doi.org/10.1109/LPT.2012.2195780>.

- [10] J. Zhang, F. Chen, B. Liu, Integrated photonic electric field sensor operating more than 26 GHz, *IEEE Microw. Wireless Compon. Lett.* 30 (2020) 1009–1012, <https://doi.org/10.1109/LMWC.2020.3019448>.
- [11] Y. Xue, Z. Ruan, L. Liu, Electrode-free photonic electric field sensor on thin film LN with high sensitivity, *Opt. Lett.* 47 (2022) 2097–2100, <https://doi.org/10.1364/OL.456750>.
- [12] C. Gutiérrez-Martínez, J. Santos-Aguilar, Electric field sensing scheme based on matched LN electro-optic retarders, *IEEE Trans. Instrum. Meas.* 50 (2008) 1362–1368, <https://doi.org/10.1109/TIM.2008.917174>.
- [13] B. Wang, R. Zeng, et al., Sensitivity improvement of integrated optical E-field sensor based on common path interferometer[C], *SENSORS, IEEE* 2012 (2012) 1–5, <https://doi.org/10.1109/ICSENS.2012.6411154>.
- [14] J. Zhang, F. Chen, B. Sun, Integrated optical E-field sensor for intense nanosecond electromagnetic pulse measurement, *IEEE Photon. Technol. Lett.* 26 (2014) 275–277, <https://doi.org/10.1109/LPT.2013.2292567>.
- [15] C. Gutiérrez-Martínez, J. Santos-Aguilar, J. Meza-Pérez, A. Morales-Díaz, Novel electric field sensing scheme using integrated optics LiNbO₃ unbalanced Mach-Zehnder interferometers and optical delay-modulation, *J. Lightwave Technol.* 35 (2017) 27–33, <https://doi.org/10.1109/JLT.2016.2630003>.
- [16] J. Zhang, L. Yang, Y. Li, Non-invasive measurement of intensive power-frequency electric field using a LN-integrated optical waveguide sensor, *IET Sci. Meas. Technol.* 15 (2020) 101–108, <https://doi.org/10.1049/smt2.12014>.
- [17] J. Zhang, F. Chen, An electro-optic sensor for measurement of intensive DC electric field, *IEEE Photon. J.* 14 (2022) 5817407, <https://doi.org/10.1109/JPHOT.2022.3153644>.
- [18] X. Li, X. Hai, G. Yao, Y. Cui, Y. Xi, P. Li, Temperature-Insensitive integrated optical electric field sensor based on asymmetric Mach-Zehnder interferometer, *IEEE Sens. J.* 23 (2023) 6963–6968, <https://doi.org/10.1109/JSEN.2023.3241964>.
- [19] Y. Shi, W. Wang, Z. Zhu, J. Yang, X. Nie, Y. Cheng, A bias controllable birefringence modulation measurement system for sub nanosecond pulsed electric field, *IEEE Trans. Instrum. Meas.* 70 (2021), <https://doi.org/10.1109/TIM.2021.3037307>.
- [20] A. Garzarella, S.B. Qadri, T.J. Wieting, D.H. Wu, R.J. Hinton, Dielectrically induced sensitivity enhancements in electrooptic field sensors, *Opt. Lett.* 32 (2007) 946–966, <https://doi.org/10.1364/OL.32.000964>.
- [21] A.F. Kip, *Fundamentals of electricity and magnetism, McGraw-Hill Series in Fundamentals of Physics: An undergraduate textbook program* (1969) 150–151.
- [22] J. Noda, M. Fukuma, S. Saito, Effect of Mg diffusion on Ti-diffused LN waveguides, *J. Appl. Phys.* 49 (1978) 3150–3154, <https://doi.org/10.1063/1.325308>.
- [23] R.J. Holmes, D.M. Smyth, Titanium diffusion into LN a function of stoichiometry, *J. Appl. Phys.* 55 (1984) 3531–3535, <https://doi.org/10.1063/1.332942>.
- [24] K. Sugii, M. Fukuma, H. Iwasaki, A study on titanium diffusion into LN waveguides by electron probe analysis and X-ray diffraction methods, *J. Mater. Sci.* 13 (1978) 523–532, <https://doi.org/10.1007/BF00541802>.
- [25] Y. Shi, W. Wang, Z. Zhu, J. Yang, X. Nie, Y. Cheng, A bias controllable birefringence modulation measurement system for subnanosecond pulsed electric field, *IEEE Trans. Instrum. Meas.* 70 (2021), <https://doi.org/10.1109/TIM.2020.3037307>.
- [26] R.V. Schmidt, I.P. Kaminow, Metal-diffused optical waveguides in LN, *Appl. Phys. Lett.* 25 (1974) 458–460, <https://doi.org/10.1063/1.1655547>.
- [27] O. Ogawa, T. Sowa, S. Ichizono, A guided-wave optical electric field sensor with improved temperature stability, *J. Lightwave Technol.* 17 (1999) 823–829, <https://doi.org/10.1109/50.762899>.
- [28] M. Yamada, Y. Nishida, K. Yamamoto, Design of low-loss Ti: LN optical waveguide with optimal waveguide width, *Jpn. J. Appl. Phys.* 45 (2006) 8343–8348, <https://doi.org/10.1143/JJAP.45.8343>.
- [29] C. Canali, C. De Bernardi, M. De Sario, A. Loffredo, G. Mazzi, S. Morasca, Effects of water vapor on refractive index profiles in Ti: LN planar waveguides, *J. Lightwave Technol.* 4 (1986) 951–955, <https://doi.org/10.1109/JLT.1986.1074820>.
- [30] S.M. Pang, S.B. Huang, Design and parameter selection of Ti: LN single mode waveguide, *Photoelectron Laser.* 4 (1993), <https://doi.org/10.16136/j.joel.1993.06.007>.
- [31] National technical committee for standardization of quantity and unit, GB/T12720-91. Power frequency electric field measurement[S], Standards Press of China, Beijing, 1991, pp.1–10.
- [32] O. Ogawa, T. Sowa, S. Ichizono, A guided-wave optical electric field sensor with improved temperature stability, *J. Lightwave Technol.* 17 (1999) 823–830, <https://doi.org/10.1109/50.762899>.